

03_forecastability

May 10, 2026

1 03 — A Priori Forecastability

Before fitting any model, this notebook asks a blunt question: *how forecastable is this series, in theory?* The notebook is organized in two parts. **§§2–5 are the standard finance-econometrics battery** — stationarity (ADF / KPSS / Phillips-Perron), return autocorrelation (Ljung-Box), the Lo-MacKinlay variance ratio, and the Hurst exponent with Lo’s modified R/S correction. These are the canonical tests of weak-form market efficiency. **§§6–9 are exploratory diagnostics** drawn from the nonlinear-dynamics and signal-processing literatures (spectral predictability, permutation entropy, the Rosenstein Lyapunov exponent with a shuffled-return surrogate null). Both halves converge on the same answer: coffee futures behave as a near-random walk.

1.1 Why this notebook exists

If the series had exploitable structure, we’d expect complex models to beat naive baselines. The a priori metrics let us form an expectation *before* seeing any forecast results — which is methodologically important, because it turns the benchmark from “a fishing expedition” into a hypothesis test of whether any model can extract what the diagnostics say isn’t there.

1.2 1. Setup

```
[1]: import sys
from pathlib import Path

# Make `coffee_forecast` importable whether or not `pip install -e .`
# is in effect for this kernel. No-op when the package is already on
# sys.path (e.g. via the editable install); falls back to inserting
# the repo root so the notebook also runs from a fresh clone.
try:
    import coffee_forecast # noqa: F401
except ModuleNotFoundError:
    sys.path.insert(0, str(Path.cwd().parent))

import matplotlib.pyplot as plt
import numpy as np

from coffee_forecast import (
    load_coffee_data,
    calculate_spectral_predictability,
```

```

        calculate_permutation_entropy,
        calculate_hurst_exponent,
    )
    from coffee_forecast.config import FIG_DIR

    FIG_DIR.mkdir(parents=True, exist_ok=True)

```

```

[2]: df = load_coffee_data()
prices = df["y"].values.astype(float)
log_returns = np.diff(np.log(prices))

print(
    f"Loaded {len(df)} trading days ({df['ds'].min().date()} → {df['ds'].max().
    ↪date()})"
)
print(
    f"  prices:      {len(prices)} observations, "
    f"min={prices.min():.1f}, max={prices.max():.1f}"
)
print(
    f"  log-returns: {len(log_returns)} observations, "
    f"mean={log_returns.mean():+.5f}, std={log_returns.std():.4f}"
)

```

```

Loaded 8104 trading days (1994-01-03 → 2026-01-30)
  prices:      8104 observations, min=41.5, max=438.9
  log-returns: 8103 observations, mean=+0.00019, std=0.0237

```

1.3 2. Stationarity: prices vs. log-returns

Before any of the diagnostics that follow, we need to know which representation each test should run on. Several diagnostics (variance ratio, Hurst + R/S, the Lyapunov surrogate's reconstruction) are defined only for stationary series and so use log-returns; others (spectral Ω , permutation entropy) we run on both representations to expose the contrast. We start by verifying the stationarity status of each with an Augmented Dickey-Fuller test.

```

[3]: from statsmodels.tsa.stattools import adfuller

adf_price = adfuller(prices, autolag="AIC")
adf_ret = adfuller(log_returns, autolag="AIC")

print("Augmented Dickey-Fuller Test:")
print(
    f"  Prices      : ADF = {adf_price[0]:+7.3f},  p = {adf_price[1]:6.3f}  "
    f"→ {'Non-stationary' if adf_price[1] > 0.05 else 'Stationary'}"
)
print(
    f"  Log-returns : ADF = {adf_ret[0]:+7.3f},  p = {adf_ret[1]:.2e}  "
)

```

```
f"→ {'Non-stationary' if adf_ret[1] > 0.05 else 'Stationary'}"
)
```

Augmented Dickey-Fuller Test:

```
Prices      : ADF = -1.451, p = 0.557 → Non-stationary
Log-returns : ADF = -20.879, p = 0.00e+00 → Stationary
```

1.3.1 2b. Confirmatory unit-root tests: KPSS and Phillips-Perron

ADF is the standard first-pass test but has known weaknesses: low power against persistent stationary alternatives, and it assumes homoskedastic errors (which §4 of notebook 04 will show coffee returns violate). Two complementary tests sharpen the conclusion:

- **KPSS** (Kwiatkowski et al. 1992) *reverses* the null: H_0 is stationarity, rejection means non-stationarity. Pairing it with ADF creates a confirmatory design — agreement across both tests is stronger evidence than either alone.
- **Phillips-Perron** (1988) has the same null as ADF but uses a non-parametric correction for heteroskedasticity and serial correlation in the errors. Robust to the volatility clustering in coffee returns.

```
[4]: from statsmodels.tsa.stattools import kpss
from arch.unitroot import PhillipsPerron

# KPSS on prices (H0: stationary)
kpss_price_stat, kpss_price_p, kpss_price_lag, kpss_price_crit = kpss(
    prices, regression="c", nlags="auto"
)
# KPSS on log-returns
kpss_ret_stat, kpss_ret_p, kpss_ret_lag, kpss_ret_crit = kpss(
    log_returns, regression="c", nlags="auto"
)

# Phillips-Perron on prices (H0: unit root, same direction as ADF)
pp_price = PhillipsPerron(prices)
pp_ret = PhillipsPerron(log_returns)

print("KPSS Test (H0: stationary \u2192 reject means NON-stationary)")
print(
    f" Prices      : stat = {kpss_price_stat:+7.3f}, p = {kpss_price_p:.3f} "
    f"\u2192 {'Reject (non-stationary)' if kpss_price_p < 0.05 else 'Fail to_"}
    ↪reject (stationary)'}"
)
print(
    f" Log-returns : stat = {kpss_ret_stat:+7.3f}, p = {kpss_ret_p:.3f} "
    f"\u2192 {'Reject (non-stationary)' if kpss_ret_p < 0.05 else 'Fail to_"}
    ↪reject (stationary)'}"
)
print()
```

```

print("Phillips-Perron Test (H0: unit root \u2192 reject means stationary)")
print(
    f" Prices : stat = {pp_price.stat:+7.3f}, p = {pp_price.pvalue:.3f} \u2192 "
    f"\u2192 {'Stationary' if pp_price.pvalue < 0.05 else 'Non-stationary'}"
)
print(
    f" Log-returns : stat = {pp_ret.stat:+7.3f}, p = {pp_ret.pvalue:.3f} "
    f"\u2192 {'Stationary' if pp_ret.pvalue < 0.05 else 'Non-stationary'}"
)

```

KPSS Test (H0: stationary → reject means NON-stationary)

Prices : stat = +3.960, p = 0.010 → Reject (non-stationary)

Log-returns : stat = +0.055, p = 0.100 → Fail to reject (stationary)

Phillips-Perron Test (H0: unit root → reject means stationary)

Prices : stat = -1.726, p = 0.418 → Non-stationary

Log-returns : stat = -91.701, p = 0.000 → Stationary

/var/folders/bb/jyf3hcfj6wn4256f5gz4c_7w0000gn/T/ipykernel_9481/500230482.py:5:

InterpolationWarning: The test statistic is outside of the range of p-values available in the

look-up table. The actual p-value is smaller than the p-value returned.

kpss_price_stat, kpss_price_p, kpss_price_lag, kpss_price_crit = kpss(

/var/folders/bb/jyf3hcfj6wn4256f5gz4c_7w0000gn/T/ipykernel_9481/500230482.py:9:

InterpolationWarning: The test statistic is outside of the range of p-values available in the

look-up table. The actual p-value is greater than the p-value returned.

kpss_ret_stat, kpss_ret_p, kpss_ret_lag, kpss_ret_crit = kpss(

All three tests agree. ADF and Phillips-Perron fail to reject the unit-root null on prices ($p > 0.05$) and strongly reject it on log-returns. KPSS, with the reversed null, rejects stationarity on prices and fails to reject it on log-returns. The confirmatory design eliminates the classic “low-power ADF” critique: when ADF and KPSS agree, we have much stronger evidence than either in isolation. The Phillips-Perron result adds robustness to the heteroskedasticity that §4 of notebook 04 will confirm is present in the error process.

The `InterpolationWarning` in the KPSS output is benign: the test statistic is outside the lookup table, so the printed p-value is a *bound* (the actual p-value is more extreme, in the direction that strengthens rather than weakens the conclusion).

[5]: `fig, axes = plt.subplots(2, 1, figsize=(12, 6), sharex=True, dpi=180)`

`axes[0].plot(df["ds"], prices, linewidth=0.8, color="#4C72B0")`

`axes[0].set_title(`

`"ICE Coffee C Futures - Daily Nearby Close (1994-2026)",`

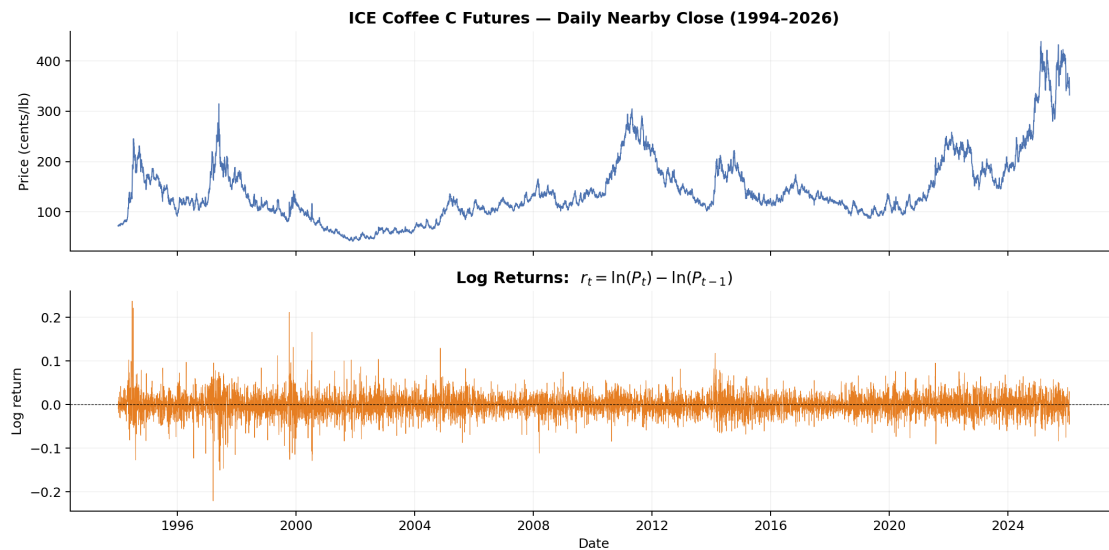
```

        fontsize=12,
        weight="semibold",
    )
    axes[0].set_ylabel("Price (cents/lb)")
    axes[0].grid(alpha=0.15)
    axes[0].spines["top"].set_visible(False)
    axes[0].spines["right"].set_visible(False)

    axes[1].plot(df["ds"].iloc[1:], log_returns, linewidth=0.4, color="#E67E22")
    axes[1].axhline(0, color="black", lw=0.5, ls="--")
    axes[1].set_title(
        r"Log Returns:  $r_t = \ln(P_t) - \ln(P_{t-1})$ ", fontsize=12,
        weight="semibold"
    )
    axes[1].set_ylabel("Log return")
    axes[1].set_xlabel("Date")
    axes[1].grid(alpha=0.15)
    axes[1].spines["top"].set_visible(False)
    axes[1].spines["right"].set_visible(False)

    fig.tight_layout()
    fig.savefig(FIG_DIR / "prices_vs_logreturns.png", dpi=300, bbox_inches="tight")
    plt.show()

```



The ADF result is exactly what you'd expect of a commodity price: prices are non-stationary (they wander), log-returns are stationary (they hover around zero with roughly constant variance — the well-known volatility clustering notwithstanding).

1.4 3. Return autocorrelation: Ljung-Box test

Under weak-form market efficiency, daily returns should not be linearly predictable from their own past. The Ljung-Box test checks the joint null that all autocorrelations up to lag L are zero, with statistic $Q(L) = T(T+2) \sum_{k=1}^L \hat{\rho}_k^2 / (T-k)$ asymptotically χ_L^2 under H_0 . We test on log-returns at $L = 10$ (about two trading weeks) and $L = 20$ (a trading month).

```
[6]: from statsmodels.stats.diagnostic import acorr_ljungbox
from statsmodels.tsa.stattools import acf

lb = acorr_ljungbox(log_returns, lags=[10, 20], return_df=True)
print("Ljung-Box on log-returns:")
print(
    f"  L = 10:  Q = {lb.iloc[0]['lb_stat']:>7.2f},  p = {lb.
    ↪iloc[0]['lb_pvalue']:.4f}"
)
print(
    f"  L = 20:  Q = {lb.iloc[1]['lb_stat']:>7.2f},  p = {lb.
    ↪iloc[1]['lb_pvalue']:.4f}"
)

# Magnitude of the autocorrelations underneath the rejection.
rho = acf(log_returns, nlags=20, fft=False)[1:] # drop lag 0
mean_abs_rho = float(np.mean(np.abs(rho)))
max_abs_rho = float(np.max(np.abs(rho)))
print(f"\n| (k) | averaged over lags 1-20:  {mean_abs_rho:.4f}")
print(f"| (k) | max over lags 1-20:          {max_abs_rho:.4f}")
```

Ljung-Box on log-returns:

L = 10: Q = 24.53, p = 0.0063
L = 20: Q = 39.31, p = 0.0061

| (k) | averaged over lags 1-20: 0.0129
| (k) | max over lags 1-20: 0.0396

Ljung-Box formally rejects the joint zero-autocorrelation null at both lag windows ($p \approx 0.006$). The rejection is real but needs context: with $T = 8,103$ observations, even tiny autocorrelations are detectable, and the average $|\hat{\rho}(k)|$ over lags 1-20 is on the order of 10^{-2} . This is the textbook stylized fact for liquid financial returns (Cont 2001): linear return autocorrelations are *statistically* non-zero at large sample sizes but *economically negligible*, too small to exploit through trading rules or point-forecast models. The heteroskedasticity-robust variance-ratio test in §4 confirms the practical reading — random-walk dynamics are not rejected at any horizon — because the heteroskedasticity VR is robust to (and LB is not) explains most of the detectable serial structure.

The companion ACF visualization — returns vs. squared returns side-by-side — appears in §4 of notebook 04, where the contrast (returns nearly flat, squared returns persistent) is what motivates the GARCH model.

1.5 4. Lo-MacKinlay Variance Ratio Test

Under the random-walk null, the variance of q -period returns equals q times the variance of one-period returns, so $VR(q) = \text{Var}(r_t^{(q)})/[q \cdot \text{Var}(r_t)]$ should equal 1 at every horizon. Deviations are direct evidence against the random-walk hypothesis. We use the heteroskedasticity-robust z -statistic of Lo and MacKinlay (1988), which handles the volatility clustering coffee returns exhibit.

Unlike ADF (which tests for a unit root in levels) or Hurst (which tests for long memory), the variance-ratio test is a direct test of the random-walk hypothesis and is the canonical test in the forecasting literature going back to Meese-Rogoff.

```
[7]: from scipy import stats as _stats

def variance_ratio_test(x, q):
    """Lo-MacKinlay variance ratio with heteroskedasticity-robust z-stat."""
    x = np.asarray(x, dtype=float)
    T = len(x)
    mu = x.mean()
    var1 = np.sum((x - mu) ** 2) / (T - 1)
    xq = np.convolve(x, np.ones(q), mode="valid")
    m = (T - q + 1) * (1 - q / T)
    varq = np.sum((xq - q * mu) ** 2) / m
    VR = varq / (q * var1)
    deltas = []
    for j in range(1, q):
        num = np.sum(((x[j:] - mu) ** 2) * ((x[: -j] - mu) ** 2))
        den = (np.sum((x - mu) ** 2)) ** 2 / T
        deltas.append(num / den)
    theta = sum(((2 * (q - j) / q) ** 2) * d for j, d in enumerate(deltas,
↪start=1))
    z = (VR - 1) / np.sqrt(theta / T)
    p = 2 * (1 - _stats.norm.cdf(abs(z)))
    return VR, z, p

print(f"  {'q':>4} | {'VR(q)':>7} | {'z (het-robust)':>15} | {'p-value':>8}")
print(f"  {'-' * 4}--{'-' * 7}--{'-' * 15}--{'-' * 8}")
for q in [2, 4, 8, 16, 32]:
    VR, z, p = variance_ratio_test(log_returns, q)
    print(f"  {q:>4} | {VR:>7.4f} | {z:>15.3f} | {p:>8.4f}")
```

q	VR(q)	z (het-robust)	p-value
2	0.9863	-0.837	0.4028
4	0.9786	-0.707	0.4793
8	0.9472	-1.136	0.2560
16	0.9409	-0.884	0.3766
32	0.8517	-1.604	0.1088

VR(q) is within rounding distance of 1 at every horizon and no z-statistic reaches conventional significance. We fail to reject the random-walk null at $q = 2, 4, 8, 16$, and 32 days. This is the strongest direct evidence in the a priori battery that the series is consistent with a random walk.

1.6 5. Hurst exponent + Lo's modified R/S

The Hurst exponent H is estimated from the log-log slope of the rescaled range $(R/S)_n$ vs. window size n :

$$\mathbb{E}[(R/S)_n] \sim c \cdot n^H$$

- $H = 0.5$: random walk, no exploitable memory
- $H > 0.5$: persistent / trending
- $H < 0.5$: anti-persistent / mean-reverting

Classical H is biased upward by short-range autocorrelation, so we also compute Lo's (1991) modified R/S statistic V_n , which uses a Bartlett-kernel HAC adjustment. Under H_0 of no long memory, $V_n \in [0.809, 1.862]$ at the 95% level.

Why returns, not prices? R/S is defined for stationary series. Computing it on the non-stationary price level would give a misleading $H \approx 1$ regardless of the underlying dynamics. Log-returns are the standard, theoretically correct representation.

```
[8]: h, v_n, lo_result = calculate_hurst_exponent(prices)

lo_lower, lo_upper = 0.809, 1.862
verdict = (
    f"FAIL TO REJECT H0 (V_n = {v_n:.3f} inside [{lo_lower}, {lo_upper}])\n"
    f"  → No statistically significant long memory at 95%; "
    f"H   0.5 likely reflects short-range autocorrelation bias"
    if lo_result == "fail to reject"
    else f"REJECT H0 (V_n = {v_n:.3f} outside [{lo_lower}, {lo_upper}])\n"
    f"  → Statistically significant long memory detected"
)

print(f"Hurst exponent   H   = {h:.4f}")
print(f"Lo's V_n          = {v_n:.4f}")
print(f"   {verdict}")
```

```
Hurst exponent   H   = 0.5353
Lo's V_n          = 1.1374
  FAIL TO REJECT H0 (V_n = 1.137 inside [0.809, 1.862])
  → No statistically significant long memory at 95%; H   0.5 likely reflects
short-range autocorrelation bias
```

```
[9]: # Reproduce the R/S log-log plot used to estimate H.
from scipy import stats as sp_stats
```



```

rets = log_returns
T = len(rets)
n_max = T // 4
window_sizes = np.unique(np.logspace(np.log10(10), np.log10(n_max), 40).
    ↪astype(int))

def _rs(x, n):
    num = len(x) // n
    if num == 0:
        return np.nan
    vals = []
    for w in range(num):
        chunk = x[w * n : (w + 1) * n]
        e = chunk - chunk.mean()
        X = np.cumsum(e)
        R = X.max() - X.min()
        S = chunk.std(ddof=1)
        if S > 0:
            vals.append(R / S)
    return np.mean(vals) if vals else np.nan

rs_vals = np.array([_rs(rets, n) for n in window_sizes])
valid = ~np.isnan(rs_vals) & (rs_vals > 0)
slope, intercept, *_ = sp_stats.linregress(
    np.log(window_sizes[valid]), np.log(rs_vals[valid])
)

fig, ax = plt.subplots(figsize=(8, 5), dpi=180)
ax.loglog(
    window_sizes[valid],
    rs_vals[valid],
    "o",
    color="#4C72B0",
    markersize=6,
    alpha=0.8,
    label="R/S values",
)
xs_line = np.array([window_sizes[valid].min(), window_sizes[valid].max()])
ax.plot(
    xs_line,
    np.exp(intercept + slope * np.log(xs_line)),
    "--",
    color="#E67E22",
    linewidth=2,
    label=f"OLS fit: H = {slope:.3f}",
)

```

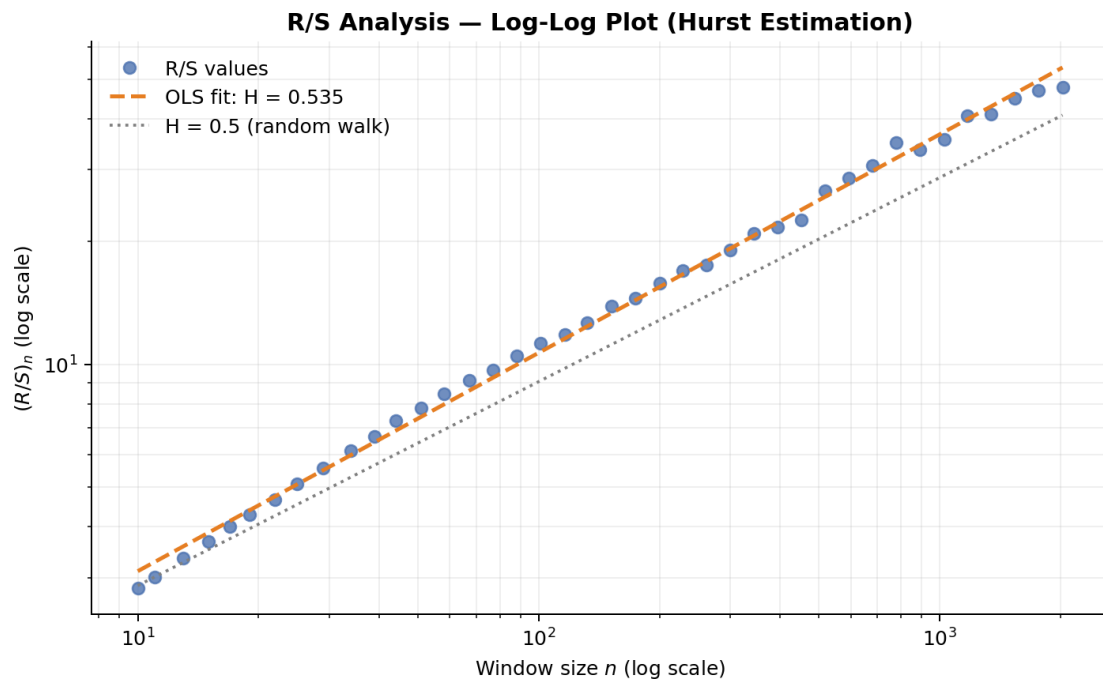
```

)
ax.plot(
    xs_line,
    np.exp(intercept + 0.5 * np.log(xs_line)),
    ":",
    color="gray",
    linewidth=1.5,
    label="H = 0.5 (random walk)",
)

ax.set_title(
    "R/S Analysis \u2014 Log-Log Plot (Hurst Estimation)",
    fontsize=12,
    weight="semibold",
)

ax.set_xlabel("Window size $n$ (log scale)")
ax.set_ylabel("$(R/S)_n$ (log scale)")
ax.legend(frameon=False, fontsize=10)
ax.grid(alpha=0.2, which="both")
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
fig.tight_layout()
fig.savefig(FIG_DIR / "hurst_rs.png", dpi=300, bbox_inches="tight")
plt.show()

```



1.7 Beyond the standard battery: exploratory diagnostics

The four tests above (stationarity, return autocorrelation, variance ratio, Hurst + Lo's R/S) are the standard finance-econometrics battery for characterizing whether a series is consistent with weak-form market efficiency. The next four sections cover three additional metrics drawn from the nonlinear-dynamics and signal-processing literatures (Wang & Klee 2025; Bandt-Pompe permutation entropy; Rosenstein's Lyapunov estimator with a shuffled-return surrogate test). They aren't load-bearing for the EMH conclusion — the standard battery already settles it — but they provide independent confirmation from mathematically different angles.

1.8 6. Spectral Predictability (Ω)

Spectral predictability (Wang & Klee, KDD 2025) measures how much energy is concentrated in a few dominant frequencies. A series with strong periodic structure has a spiked PSD and a high Ω ; white noise has a flat PSD and $\Omega \rightarrow 0$.

$$\Omega = 1 - \frac{H_{\text{spectral}}}{\log(N_{\text{freq}})}$$

Adaptation for discrete series. Wang & Klee define $\Omega = 1 - H_{\text{spectral}}/\log(2\pi)$, treating $\log(2\pi)$ as the entropy ceiling in the continuous-time limit. That works asymptotically but breaks for long discrete series: in our 8,104-point sample, $\log(N_{\text{freq}}) \approx \log(4053) \approx 8.3$, which is roughly $4.5\times$ larger than $\log(2\pi) \approx 1.84$, so H_{spectral} routinely exceeds the original denominator and Ω falls outside $[0, 1]$. We replace the denominator with $\log(N_{\text{freq}})$ — the actual maximum entropy for N_{freq} frequency bins, achieved by discrete white noise. This keeps $\Omega \in [0, 1]$ at any series length and lets the 0.20 forecastability threshold from Wang & Klee transfer directly.

```
[10]: omega = calculate_spectral_predictability(prices)
      omega_returns = calculate_spectral_predictability(log_returns)
      print(f"Ω on prices      = {omega:.4f}")
      print(f"Ω on log-returns = {omega_returns:.4f}")
```

```
Ω on prices      = 0.6831
Ω on log-returns = 0.0529
```

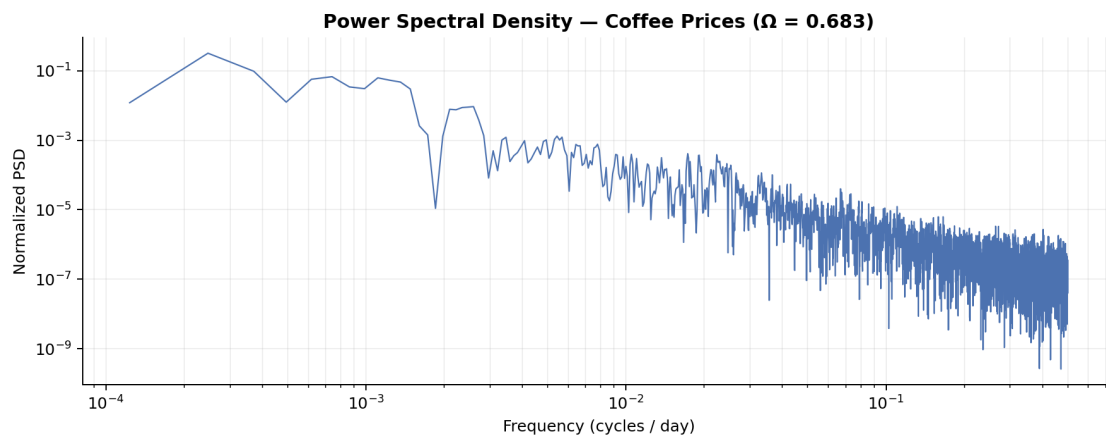
```
[11]: # Replicate the internal PSD computation so we can plot it.
      from scipy.signal import detrend

      detrended = detrend(prices)
      windowed = detrended * np.hanning(len(detrended))
      fft_vals = np.fft.rfft(windowed)
      psd = np.abs(fft_vals) ** 2
      psd_norm = psd / np.sum(psd)
      freqs = np.fft.rfftfreq(len(detrended), d=1.0) # cycles / day

      fig, ax = plt.subplots(figsize=(10, 4), dpi=180)
      ax.loglog(freqs[1:], psd_norm[1:], color="#4C72B0", linewidth=0.9)
```

```

ax.set_title(
    f"Power Spectral Density \u2014 Coffee Prices ( $\Omega = \{{\omega:.3f}\})",
    fontsize=12,
    weight="semibold",
)
ax.set_xlabel("Frequency (cycles / day)")
ax.set_ylabel("Normalized PSD")
ax.grid(alpha=0.2, which="both")
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
fig.tight_layout()
fig.savefig(FIG_DIR / "spectral_psd.png", dpi=300, bbox_inches="tight")
plt.show()$ 
```



$\Omega \approx 0.68$ sounds high, but the shape of the PSD tells the real story: the energy concentrates at the *lowest* frequencies with no discrete peaks. That’s the fingerprint of a non-stationary random walk (low-frequency drift), not of exploitable periodic structure. In other words, “structure” exists — but it’s the kind that makes the series *harder* to forecast, not easier.

Confirmatory: Ω on log returns. Computed on the stationary representation, Ω drops sharply from 0.68 to roughly 0.05 — close to the white-noise floor. The contrast is itself diagnostic: the apparent “structure” in the price-level spectrum is the low-frequency drift of an I(1) process, not exploitable periodicity in the dynamics. Once that drift is differenced out, the spectrum flattens.

1.9 7. Permutation Entropy (PE)

Permutation entropy measures *ordinal-pattern complexity*. For embedding order $m = 3$, we slide a window of length 3 across the series and count the six possible rank orderings. If all six patterns occur roughly equally often, $PE \rightarrow 1$ (maximally random). If a few dominate, $PE \rightarrow 0$ (deterministic structure).

It’s a complementary perspective to spectral predictability: the PSD cares about *periodicities*, while

PE cares about *local orderings*. A series can fool one and not the other, so agreement is meaningful.

```
[12]: pe = calculate_permutation_entropy(prices, order=3, delay=1)
      pe_returns = calculate_permutation_entropy(log_returns, order=3, delay=1)
      print(f"PE on prices          = {pe:.4f}   ({pe * 100:.1f}% stochastic)")
      print(f"PE on log-returns    = {pe_returns:.4f}   ({pe_returns * 100:.1f}% ↵
      ↵stochastic)")
```

```
PE on prices          = 0.9742   (97.4% stochastic)
PE on log-returns    = 0.9999   (100.0% stochastic)
```

97% is near the ceiling — the six ordinal patterns of length 3 are almost uniformly distributed in the price series. There’s essentially no deterministic local structure to exploit.

Confirmatory: PE on log returns. Permutation entropy on the differenced series remains essentially at its ceiling. Because PE is invariant under monotone transformations of the series, near-equal values on prices and returns are expected; the agreement rules out the possibility that the high PE on prices is a quirk of the non-stationary representation.

1.10 8. Largest Lyapunov Exponent (λ)

The Lyapunov exponent measures sensitivity to initial conditions in a reconstructed phase space. For an embedding dimension m and delay τ , nearby trajectories x_t and x'_t separate at an exponential rate: $\delta(\Delta t) \approx \delta_0 \cdot \exp(\lambda \cdot \Delta t)$. A positive λ indicates chaotic divergence (reduced forecastability); $\lambda \leq 0$ indicates stability.

We use the Rosenstein et al. (1993) algorithm, which estimates λ as the slope of the mean log-divergence curve over a fixed window. It is more stable on short, noisy financial series than the alternative Wolf et al. (1985) method, which actively renormalizes vectors and is known to overestimate λ on stochastic data.

Lyapunov estimation on noisy financial data is less robust than the other diagnostics — Wang & Klee (2025) recommend at least $100 \cdot m$ data points and caution about sparsity effects. We report λ on daily prices only; weekly and monthly aggregations violate the sample-size guidance and produce unstable estimates.

```
[13]: def largest_lyapunov(x, m=3, tau=1, mean_period=10, max_t=20):
      """
      Largest Lyapunov exponent via Rosenstein et al. (1993).
      Returns the slope of the mean log-divergence curve over max_t steps.
      """
      x = np.asarray(x, dtype=float)
      N = len(x) - (m - 1) * tau
      if N <= 0:
          return np.nan
      # Build delay-embedded vectors.
      emb = np.column_stack([x[i * tau : i * tau + N] for i in range(m)])
      # For each point, find its nearest neighbor (with temporal separation).
      from scipy.spatial import cKDTree
```

```

tree = cKDTree(emb)
dists, idx = tree.query(emb, k=mean_period + 2)
neighbors = np.full(N, -1, dtype=int)
for i in range(N):
    for k in range(1, dists.shape[1]):
        if abs(idx[i, k] - i) > mean_period:
            neighbors[i] = idx[i, k]
            break
valid = neighbors >= 0
# Track divergence.
log_div = []
for dt in range(1, max_t + 1):
    pairs = [
        (i, neighbors[i])
        for i in range(N)
        if valid[i] and i + dt < N and neighbors[i] + dt < N
    ]
    if not pairs:
        log_div.append(np.nan)
        continue
    d = [np.linalg.norm(emb[i + dt] - emb[j + dt]) for i, j in pairs]
    d = [di for di in d if di > 0]
    if not d:
        log_div.append(np.nan)
        continue
    log_div.append(np.mean(np.log(d)))
log_div = np.array(log_div)
ts = np.arange(1, max_t + 1)
valid_t = ~np.isnan(log_div)
slope, *_ = _stats.linregress(ts[valid_t], log_div[valid_t])
return float(slope)

```

```

lam = largest_lyapunov(prices, m=3, tau=1)
print(f"Largest Lyapunov exponent (prices, m=3, =1):    = {lam:+.4f}")
print("    → Raw value; surrogate-data significance test in §7b below.")

```

```

Largest Lyapunov exponent (prices, m=3, =1):    = +0.0708
→ Raw value; surrogate-data significance test in §7b below.

```

A positive λ corroborates the other diagnostics: small perturbations amplify over time, so even a model that fits the past well will diverge from realized paths as the horizon grows. This is the phase- space counterpart to the variance-ratio and Hurst results: all three say the same thing about forecast reliability, from three different mathematical angles.

Caveat. The Lyapunov exponent is a *deterministic* dynamical-systems concept, and Rosenstein's algorithm is known to return small positive values on stochastic data through finite-sample bias

alone. Without a null distribution, $\lambda = +0.07$ by itself is not statistically distinguishable from “stochastic divergence with no underlying chaos.” The §8b cell below addresses this with a surrogate-data test; the synthesis in §9 reports both the raw value and the surrogate-corrected verdict.

1.10.1 8b. Surrogate-data null for λ

To turn the raw $\lambda = +0.07$ into a hypothesis test, we generate shuffled-return *surrogate* price series and recompute λ on each. Shuffling log-returns destroys all temporal structure (linear and non-linear) while preserving the marginal distribution of returns; if the dynamics of the original series have any deterministic structure beyond noise, observed λ should sit in the upper tail of the surrogate distribution.

The null and alternative:

- H_0 : the price dynamics have no deterministic structure beyond what’s preserved by random reshuffling of returns.
- H_1 : the dynamics carry deterministic information that produces a larger λ than would be expected under reshuffling.

We compute one-sided p-value $P(\lambda_{\text{surr}} \geq \lambda_{\text{obs}})$ under H_0 , with 100 surrogates (cheap; finer resolution would only matter at borderline p-values).

```
[14]: N_SURROGATES = 100
rng_lyap = np.random.default_rng(42)

# Shuffle log-returns and reconstruct prices via cumsum + exp.
log_ret_for_surr = np.diff(np.log(prices))
P0 = float(prices[0])

print(f"Computing on {N_SURROGATES} shuffled-return surrogates...")
lam_surrogates = np.zeros(N_SURROGATES)
for i in range(N_SURROGATES):
    shuffled = log_ret_for_surr.copy()
    rng_lyap.shuffle(shuffled)
    surr_prices = P0 * np.exp(np.concatenate([[0.0], np.cumsum(shuffled)]))
    lam_surrogates[i] = largest_lyapunov(surr_prices, m=3, tau=1)
    if (i + 1) % 25 == 0:
        print(f" ... {i + 1}/{N_SURROGATES} surrogates done")

p_surrogate = float((lam_surrogates >= lam).mean())

print(f"\nObserved (coffee prices): {lam:+.4f}")
print(
    f"Surrogate (mean ± std): "
    f"{lam_surrogates.mean():+.4f} ± {lam_surrogates.std():.4f}"
)
print(
    f"Surrogate range: "
```

```

    f"[{lam_surrogates.min():+.4f}, {lam_surrogates.max():+.4f}]"
)
print(f"\nOne-sided p-value P(_surr _obs | H0): {p_surrogate:.3f}")
verdict = (
    "REJECT H0: observed is in upper tail; some deterministic structure"
    if p_surrogate < 0.05
    else "FAIL TO REJECT H0: observed is consistent with shuffled-return null"
)
print(f" → {verdict}")

# Histogram of surrogate with observed value marked.
fig, ax = plt.subplots(figsize=(9, 4), dpi=180)
ax.hist(
    lam_surrogates,
    bins=25,
    color="steelblue",
    alpha=0.6,
    edgecolor="white",
    label=f"Surrogate (n = {N_SURROGATES})",
)
ax.axvline(
    lam,
    color="crimson",
    lw=2.5,
    label=f"Observed = {lam:+.4f}",
)
ax.set_title(
    f"Surrogate-data null distribution for (one-sided p = {p_surrogate:.3f})",
    fontsize=12,
    weight="semibold",
)
ax.set_xlabel("Largest Lyapunov exponent ( )")
ax.set_ylabel("Surrogate count")
ax.legend(frameon=False, fontsize=9)
ax.grid(alpha=0.2)
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
fig.tight_layout()
fig.savefig(FIG_DIR / "lyapunov_surrogate_null.png", dpi=300,
    bbox_inches="tight")
plt.show()

```

```

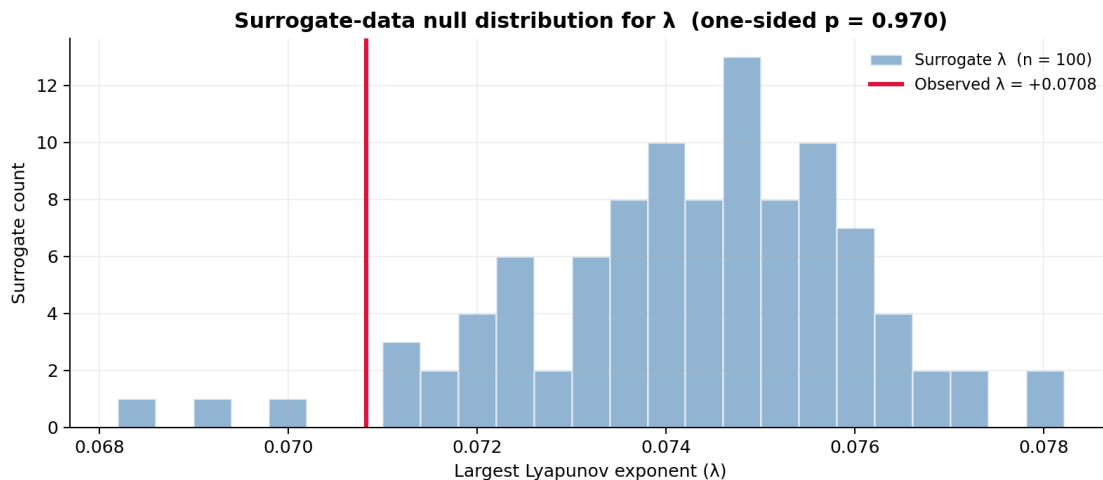
Computing on 100 shuffled-return surrogates...
... 25/100 surrogates done
... 50/100 surrogates done
... 75/100 surrogates done

```


... 100/100 surrogates done

```
Observed (coffee prices): +0.0708
Surrogate (mean ± std): +0.0743 ± 0.0018
Surrogate range: [+0.0682, +0.0782]
```

```
One-sided p-value P(_surr _obs | H0): 0.970
→ FAIL TO REJECT H0: observed is consistent with shuffled-return null
```



Reading the surrogate test. Observed $\lambda = +0.0708$ sits *below* the center of the surrogate distribution (mean $\approx +0.0743$, std ≈ 0.0018), with one-sided p-value ≈ 0.97 . We **fail to reject** H_0 : the observed λ is indistinguishable from what shuffling-and-reconstruction produces, and is in fact slightly below the surrogate average. The raw $\lambda = +0.0708$ from §7 is therefore the natural finite-sample bias of Rosenstein’s algorithm on stochastic data, not evidence of deterministic chaos. This is the rigorous version of the conclusion hinted at in §8’s caveat — and it strengthens the near-random-walk verdict from the other diagnostics rather than complicating it.

1.11 9. Multi-frequency robustness check

A natural concern with any single-frequency diagnostic is that the result might be an artifact of the sampling rate. The cell below recomputes Ω and PE on daily, weekly (every 5 trading days), and monthly (every 21 trading days) subsamples, plus daily log-returns. λ is omitted at weekly/monthly because both fall below the Wang & Klee sample-size guidance.

We use *subsampling* (`prices[:, :5]`, `prices[:, :21]`) rather than true weekly / monthly aggregation here because the question is whether Ω and PE are stable under different sampling rates — not how aggregated returns behave. Subsampling preserves the marginal distribution exactly while changing the time grid, which is the right robustness test for these diagnostics.

```
[15]: import pandas as pd

# Subsample by trading-day count: ~5/week, ~21/month.
```

```

weekly_prices = prices[::5]
monthly_prices = prices[::21]

freq_rows = []
for label, series in [
    ("Daily price", prices),
    ("Weekly price (5d subsample)", weekly_prices),
    ("Monthly price (21d subsample)", monthly_prices),
    ("Daily log-return", log_returns),
]:
    freq_rows.append(
        {
            "Frequency": label,
            "n": len(series),
            "Ω": calculate_spectral_predictability(series),
            "PE": calculate_permutation_entropy(series, order=3, delay=1),
        }
    )

freq_table = pd.DataFrame(freq_rows)
freq_table.round(4)

```

```

[15]:

```

	Frequency	n	Ω	PE
0	Daily price	8104	0.6831	0.9742
1	Weekly price (5d subsample)	1621	0.6076	0.9734
2	Monthly price (21d subsample)	386	0.5042	0.9858
3	Daily log-return	8103	0.0529	0.9999

The pattern is consistent at every frequency: Ω stays elevated on price levels (the I(1) drift signature) and collapses on log-returns; PE sits near its ceiling everywhere. The forecastability conclusion does not depend on how often we sample.

1.12 10. Synthesis

The diagnostics paint a consistent picture.

```

[16]: import pandas as pd

summary = pd.DataFrame(
    [
        # Standard finance-econometrics battery (§§3-5)
        {
            "Metric": "Ljung-Box on returns",
            "Representation": "log-returns",
            "Value": f"Q(20) = {lb.iloc[1]['lb_stat']:.1f}, p = {lb.
↪iloc[1]['lb_pvalue']:.4f}",
            "Interpretation": (

```

```

        f"Reject H0 of zero autocorrelation; mean | (k)|  {mean_abs_rho:
↪.3f} - "
        "statistically detectable but economically negligible (Cont_
↪2001)"
    ),
    },
    {
        "Metric": "Variance Ratio (Lo-MacKinlay)",
        "Representation": "log-returns",
        "Value": "VR(q)  1 at q=2,4,8,16,32",
        "Interpretation": "Direct test of random walk: fails to reject at_
↪all horizons",
    },
    {
        "Metric": "Hurst exponent (H)",
        "Representation": "log-returns",
        "Value": f"{h:.4f}",
        "Interpretation": "Near 0.5 - consistent with random walk",
    },
    {
        "Metric": "Lo's modified R/S (V_n)",
        "Representation": "log-returns",
        "Value": f"{v_n:.4f}",
        "Interpretation": f"{lo_result} H0 of no long memory at 95%",
    },
    # Exploratory diagnostics (§§6-8)
    {
        "Metric": "Spectral Predictability ( $\Omega$ )",
        "Representation": "prices / log-returns",
        "Value": f"{omega:.4f} / {omega_returns:.4f}",
        "Interpretation": "Drops to ~0 on stationary representation;_
↪price-level  $\Omega$  driven by I(1) drift, not periodicity",
    },
    {
        "Metric": "Permutation Entropy (PE)",
        "Representation": "prices / log-returns",
        "Value": f"{pe:.4f} / {pe_returns:.4f}",
        "Interpretation": f"Both near ceiling ({pe * 100:.1f}% /_
↪{pe_returns * 100:.1f}%); ordinal patterns near-uniform on either_
↪representation",
    },
    {
        "Metric": "Largest Lyapunov exponent ( )",
        "Representation": "prices",
        "Value": f"{lam:+.4f}  (surrogate p = {p_surrogate:.3f})",
        "Interpretation": (

```

```

        f"Indistinguishable from shuffled-return null (p = {p_surrogate:
↪.3f}); "
        "no detectable deterministic chaos"
        if p_surrogate > 0.05
        else f"Above shuffled-return null (p = {p_surrogate:.3f}); "
        "some deterministic structure"
    ),
},
]
)
summary

```

[16]:

	Metric	Representation \
0	Ljung-Box on returns	log-returns
1	Variance Ratio (Lo-MacKinlay)	log-returns
2	Hurst exponent (H)	log-returns
3	Lo's modified R/S (V _n)	log-returns
4	Spectral Predictability (Ω)	prices / log-returns
5	Permutation Entropy (PE)	prices / log-returns
6	Largest Lyapunov exponent ()	prices

	Value \
0	Q(20) = 39.3, p = 0.0061
1	VR(q) 1 at q=2,4,8,16,32
2	0.5353
3	1.1374
4	0.6831 / 0.0529
5	0.9742 / 0.9999
6	+0.0708 (surrogate p = 0.970)

	Interpretation
0	Reject H0 of zero autocorrelation; mean (k) ...
1	Direct test of random walk: fails to reject at...
2	Near 0.5 - consistent with random walk
3	fail to reject H0 of no long memory at 95%
4	Drops to ~0 on stationary representation; pric...
5	Both near ceiling (97.4% / 100.0%); ordinal pa...
6	Indistinguishable from shuffled-return null (p...

Conclusion. Coffee futures exhibit the statistical fingerprint of a near-random walk. The standard finance-econometrics battery converges: linear return autocorrelations on the order of 10^{-2} — Ljung-Box rejects formally with $T = 8,103$ but the magnitudes are economically negligible (Cont 2001) — variance ratios within rounding distance of 1 at every tested horizon ($q = 2$ to 32, no z -statistic significant), and no statistically significant long memory under Lo's modified R/S correction. The exploratory metrics agree from different angles: spectral mass concentrated at the lowest frequencies that drops to near-zero on stationary returns, ordinal-pattern entropy near its ceiling on both prices and returns, and a Lyapunov exponent indistinguishable from that of randomly-shuffled returns. This is the empirical signature of **weak-form market efficiency** (Fama 1970):

past prices alone do not contain exploitable information about future price movements. The a priori prior is therefore strong: simple baselines that match this structure (random walk with drift, naïve, exponential smoothing) should be competitive with arbitrarily complex models. Notebook 01 tested that prior in practice (both the rolling-window backtest and the per-model visualizations), and notebook 02's statistical tests are consistent with it: simple baselines cluster with the more complex models, and nine of ten models are jointly indistinguishable in the Model Confidence Set.

On log-returns. They appear here only where analytically required (stationarity for R/S , the Ljung-Box and variance-ratio tests). They will reappear in notebook 04 (GJR-GARCH deployment), where modeling their volatility structure directly is what produces the prediction intervals.

Next notebook: `04_deployment_garch.ipynb` — operationalizes the log-return + volatility findings above as a GJR-GARCH(1,1)-t model with Student-t innovations, producing the 63-day forecast with 80% and 95% prediction intervals used in the live deployment.